

* Software Process Models:-

① Waterfall Model:

The waterfall model is a linear sequential process where software development progresses through fixed phases:

Requirements → Design → Coding & Unit Testing → Integration → Operation & Maintenance & System testing

Advantages:

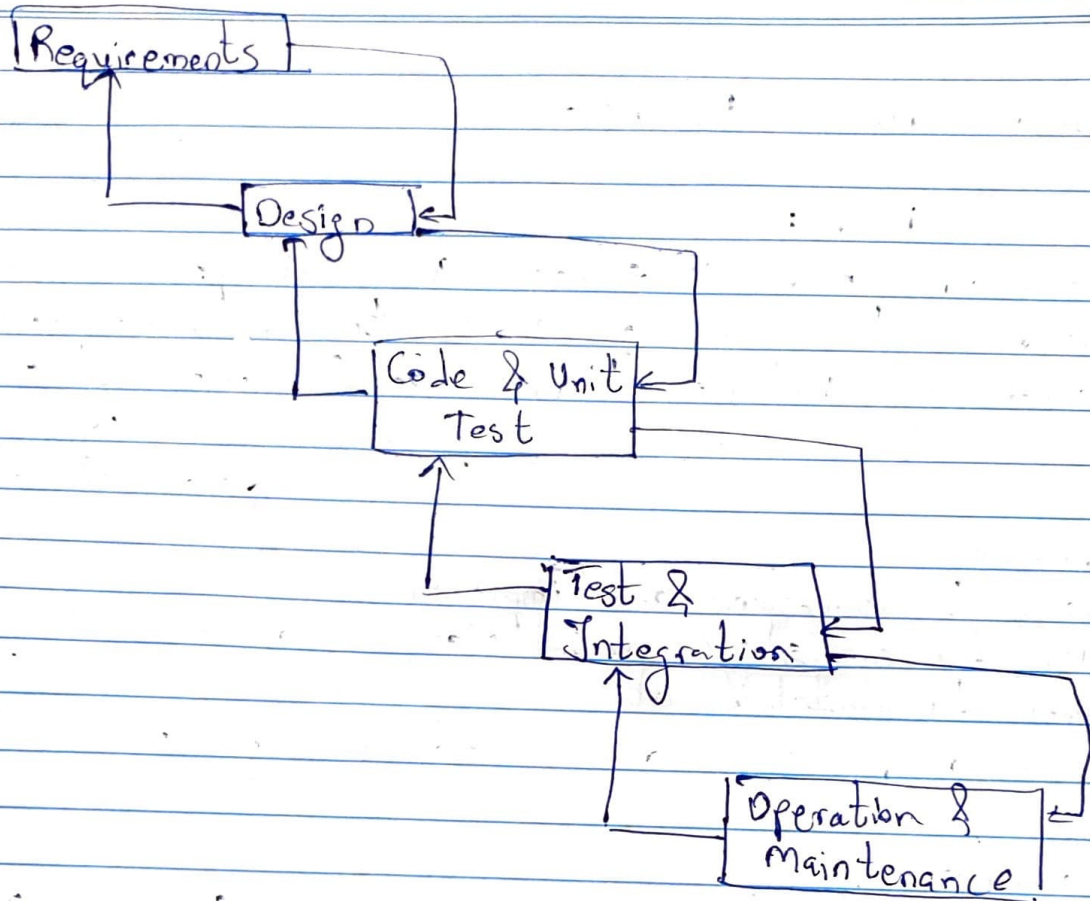
- Easy to understand and implement
- Well-defined stages with clear milestones
- Proper documentation at every stage
- Suitable for small and well-defined projects

Disadvantages:

- Very difficult to ~~accomo~~ accommodate changes once a phase is completed
- Testing is done late in the process
- Not suitable for complex or long-term projects
- No working software until late stages.

② Evolutionary Model:

In this model, software is developed in multiple versions. The initial version provides basic functional and subsequent versions are enhanced based on user feedback until the final system is delivered.



Advantages:

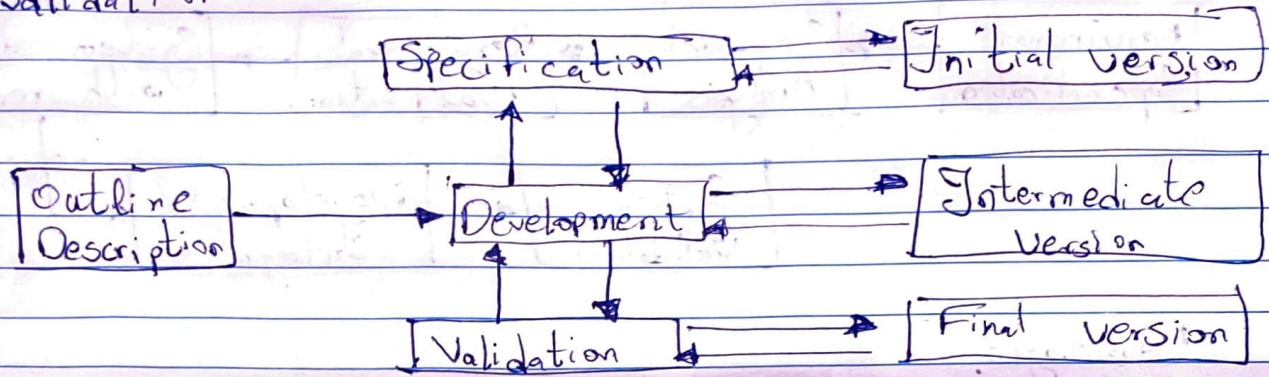
- Easy delivery of working software
- Continuous user development
- Easy to handle changing requirements
- Reduced risk of failure.

Disadvantages

- Difficult to plan and control
- May result in poor system design if changes are frequent
- Documentation may be incomplete.
- Requires experienced developers

③ Component Based Software Engineering

It focusses on building systems by reusing existing software components. The process includes requirement specification, component analysis, requirement modification, system design with reuse, development, integration and validation.



Advantages:

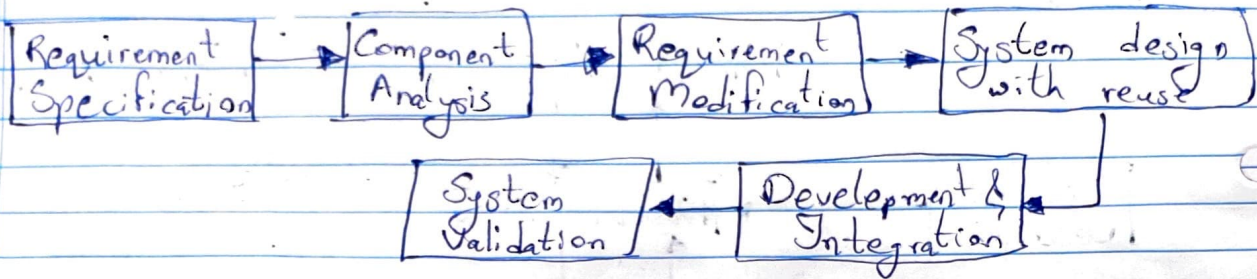
- Faster development time
- Reduced development and maintenance cost.
- High reliability due to tested components
- Encourages software reuse.

Disadvantages

- Limited control over reused components
- Compatibility and integration issues
- Performance may be affected.
- Suitable components may not always be available.

④ Incremental Model:

In the incremental model, software is developed and delivered in small increments. Each increment adds new functionality and final system is achieved after all increments are completed.



Advantages

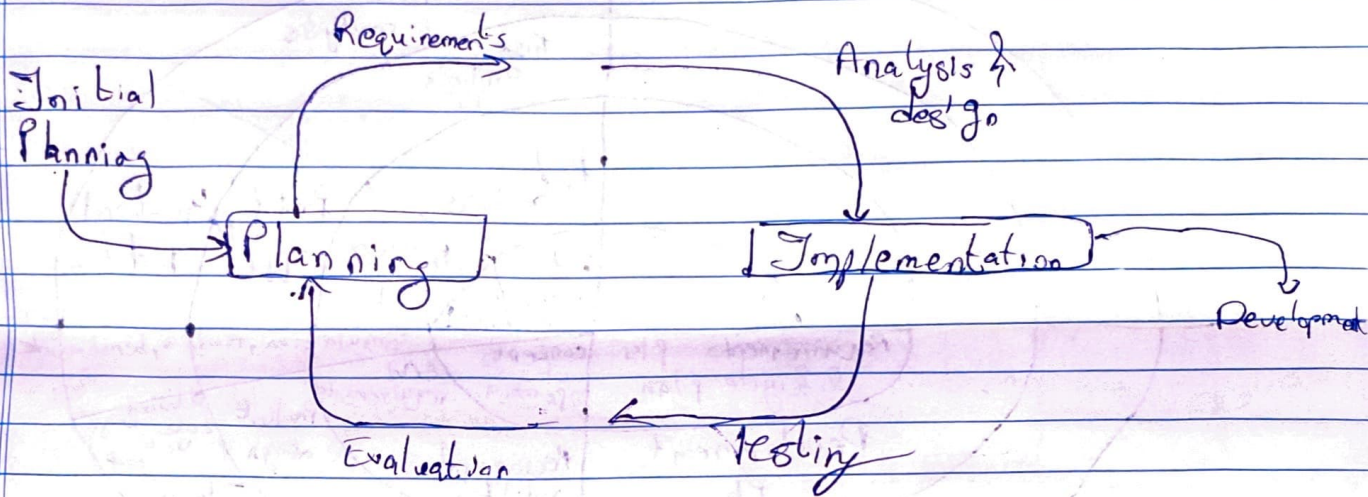
- Early delivery of partial working software
- Easier testing and debugging
- Flexible to requirement changes
- Reduced risk compared to waterfall.

Disadvantages:

- Requires good planning and design.
- Integration of increments can be complex
- Total cost may increase
- Not suitable for very small projects.

⑤ Spiral Model

The spiral model is risk driven process that combines iterative development with systematic risk analysis. Each loop of the spiral includes planning, risk analysis, development and evaluation.



Advantages:

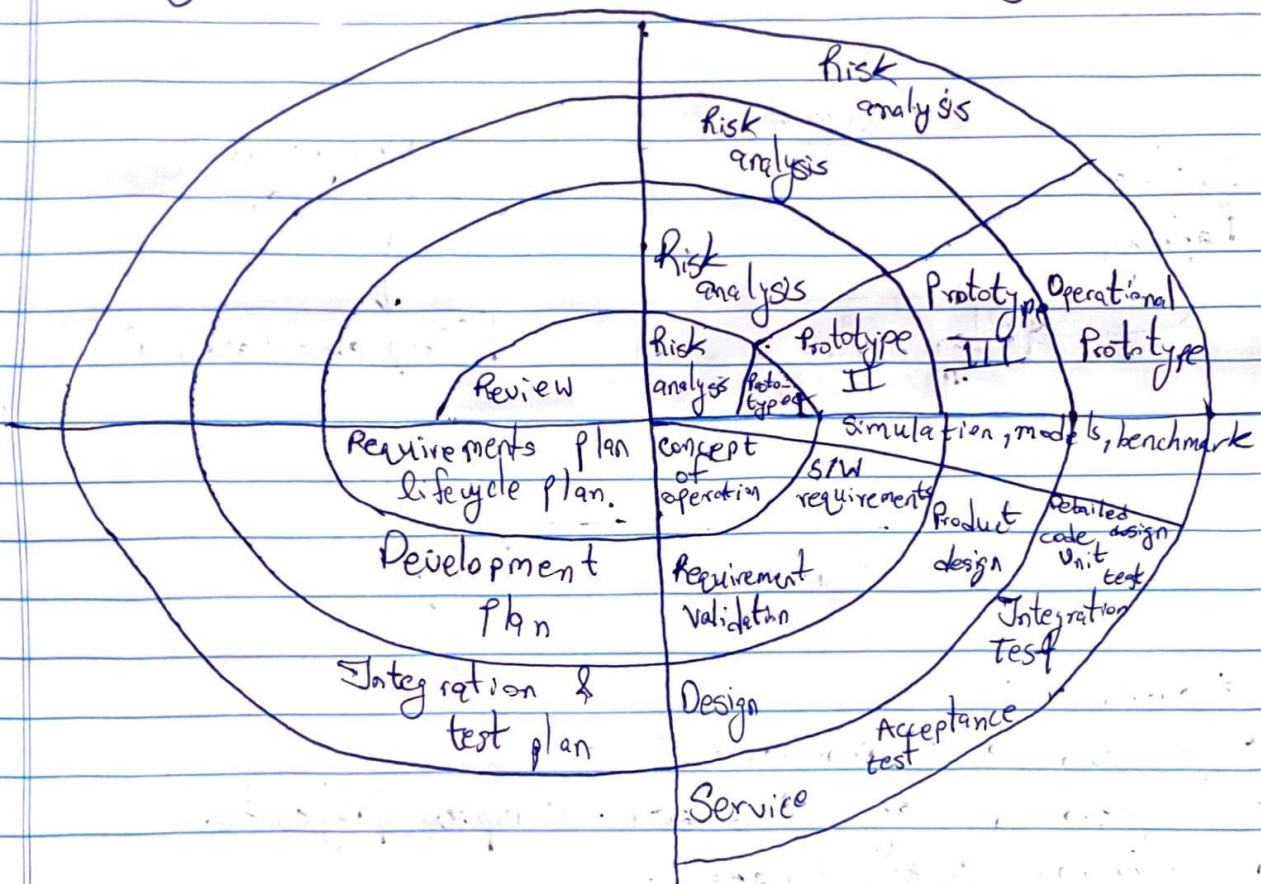
- Effective risk management
- Suitable for large and complex projects.
- Continuous user feedback.
- High quality software.

Disadvantage

- Expensive & complex to manage
- Requires expert risk assessment
- Not suitable for small projects
- Difficult to schedule.

⑥ Agile Model

Agile is an iterative and incremental approach that focuses on rapid development, continuous feedback, and close collaboration with customers. Working software is delivered frequently.



Advantage:

- Highly flexible to changes
- Improved product quality

Disadvantages:

- Less documentation
- Difficult to predict time & cost